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Apparatus for dental treatment

Abstract

An apparatus for treatment of teeth, including one or more dental treatment applicators having at least two walls and a floor, the walls and floor defining a well configured to accommodate at least one human dental arcade and an active agent. Furthermore, one or more energy-applying elements such as electrodes or light sources are located on at least one of the walls.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/470,515, filed Sep. 9, 2021, which is a continuation of U.S. patent application Ser. No. 15/814,434 (now U.S. Pat. No. 11,141,251), filed Nov. 16, 2017, which is a continuation of U.S. patent application Ser. No. 14/172,983 (now U.S. Pat. No. 9,839,500), filed Feb. 5, 2014, which claims priority from U.S. Provisional Patent Application Ser. No. 61/864,648, filed Aug. 12, 2013, the entireties of which are incorporated herein by reference.

BACKGROUND

- (1) External appearance is important to practically every person. In recent years, methods and apparatuses have been developed for various dental treatments, namely fluorination of teeth as a preventative measure against tooth decay (e.g., tooth caries) and teeth whitening for aesthetic purposes.
- (2) Whitening teeth generally refers to application of any number of chemical and/or physical (mechanical) processes, procedures, or treatments, performed singly or in combination, which result in returning, and/or providing white color to teeth.
- (3) The chemical agents most commonly used for teeth whitening are oxidizing agents such as peroxides, some of which are activated when stimulated electrically, by heat and/or by light.
- (4) Application of chemical agents has been done in the past by using a “soaking” tray applied to a dental arcade for a given period of time wherein the teeth are brought into contact with the chemical agent. The treatment demanded much patience from the subject due to the length of time required for the chemical reaction to affect the teeth and moreover, in some cases—repeated treatments.
- (5) Attempts at changing the concentration of the chemical agent and/or the amount of time of exposure to the chemical agent, for whitening teeth, have been made. For example, some techniques (methods, devices) involve use of an electrical current to stimulate ion exchange, provide a thin conductive coating on the surface of the teeth, thereby enhancing the teeth conductivity. However, some of the existing techniques are typically based on using an electrical circuit which runs through a non-intra-oral cavity body part of an individual.
- (6) A typical chemical agent most commonly used for teeth whitening is hydrogen peroxide, although other relatively strong oxidizing agents, at appropriate concentrations and conditions of oral application are also commonly used for whitening teeth. Teeth are exposed to such strong oxidizing agents, whereby the oxidizing agents oxidize, and possibly also chemically degrade, substances which discolor teeth. However, applications of techniques for whitening teeth based on use of oxidizing agents, are typically accompanied by inadvertent or unpreventable, and undesirable, exposure of non-tooth intraoral cavity components, e.g., gum and tissue exterior surfaces, to the oxidizing agents. In some cases, applications of chemical or active agents is accompanied by application of a process or a material accelerating the bleaching or whitening action of the agent.
- (7) U.S. Pat. No. 6,340,301 discloses that the bleaching/whitening agents currently used in the art can be further activated by the constant application of heat to the bleaching/whitening agent, increasing the temperature of the bleaching/whitening agent above normal body temperature (98.6 .degree. F.) with a sub-sequential and consequential increase in the bleaching/whitening reaction, pursuant to the Q10 Rule, thereby activating and accelerating the bleaching/whitening process of stained or otherwise discolored teeth.
- (8) US Patent Application Publication No. 2003/0198605 discloses a method of whitening teeth employing a combination of an oxidizing composition and an accelerator that when exposed to a

biologically safe and effective level of photoactinic light (e.g., UV light) the ability of the oxidizing compound in the whitening composition to effect rapid tooth whitening is enhanced.

(9) U.S. Pat. No. 7,775,795 to Khawaled discloses an electrochemical method and a device for treating teeth. The method requires use of an activation solution such as a metal salt solution that is applied to teeth before the treatment to increase the tooth conductivity. Following the increase of the tooth conductivity an ionizable substance is applied to teeth. Electric current flows from a source of current to the teeth through a series of contacts applied directly to the teeth, so as to ionize the ionizable substance and reduce the tooth decay.

(10) Techniques (methods, devices) currently employed could provide a uniform treatment throughout a dental arcade, but involve electric or mechanical contact with the treated teeth. Such contacts could result in certain uneasiness of the treated subject. Moreover, the techniques lack the ability to easily monitor treatment parameters, chemical quantities and rates of application applied to a dental arcade and a rapid response to gum irritation and subject discomfort should such occur.

BRIEF SUMMARY

(11) The present disclosure seeks to provide dental treatment applicators that can be fitted on either one of the upper and lower human dental arcades.

(12) There are thus provided dental treatment applicators that include two parts—a portable unit and a dental treatment tray. The tray could be designed to be paired with a same or different dental treatment tray so that both dental treatment trays can be fitted at the same time (concurrently) onto their respective upper and lower dental arcades thereby treating both dental arcades concurrently and shortening treatment time.

(13) The present disclosure also seeks to provide applicators for teeth whitening treatment that are operative to apply teeth whitening treatment.

(14) There is thus provided a dental treatment applicator having one or more energy-applying elements that can be configured to accelerate the teeth whitening activity of teeth whitening agents applied to a portion of or the full dental arcade. Such energy-applying elements could apply an electric current to active or teeth whitening agent, emit in course of treatment light energy such as, for example, light emitting diodes (LED) and apply acoustic energy causing mechanical vibrations or magnetic fields.

(15) In accordance with other examples, the energy-applying elements can be piezoelectric elements causing desired vibrations and/or a permanent or electromagnetic elements forming a magnetic field.

(16) In accordance with one example, there is also provided a dental treatment applicator, and in particular a dental tray of the applicator, that can include electrodes configured to apply electric current to a whitening agent with a suitable electric conductivity. The whitening agent could be in a liquid or gel aggregate state. The electrodes could be immersed into a whitening or active agent with a suitable electric conductivity loaded into the tray. The electrodes could be adhered to walls of the tray by a suitable biocompatible adhesive or embedded inside the walls so as to have an exposed surface to create an electric current in the whitening or active agent with a suitable electric conductivity and wherein the exposed surface does not contact the teeth.

(17) The DC-current flowing through the whitening agent, for example such a whitening gel or fluid with a suitable conductivity could also cause an oxidation reaction and a reduction reaction which is commonly referred to as a 'redox' reaction. The redox reaction involves an oxidizing agent suitable for use in teeth whitening procedure.

(18) In accordance with another example the dental treatment tray of the dental treatment applicators can also include piezoelectric elements that could protrude from the dental treatment tray walls so that to come into contact with the gums when dental treatment applicator is applied to the dental arcade.

(19) When activated, the piezoelectric elements can vibrate thus massaging the gums without the abrasive action commonly applied by apparatuses such as a tooth brush. The massaging effect can

temporarily alleviate any discomfort or irritation brought about by the active agents, which could be in liquid or gel aggregate state, and/or oxidation process.

(20) In accordance with another example, there is thus also provided a dental treatment applicator including a dental treatment tray with a well accommodating a dental arcade and active or whitening agent and piezoelectric elements that can be located so that not to come into contact with the gums when the dental treatment dental treatment applicator is applied to the dental arcade so that when activated, the piezoelectric elements can vibrate and assist in uniform distribution of the active agents throughout the well.

(21) The present disclosure also seeks to provide dental treatment applicators that are operative to apply calculus removal treatment using, for example, piezoelectric elements.

(22) In accordance with another example, there is thus also provided a dental treatment tray of a dental treatment applicator including piezoelectric elements that can affect acoustic-mechanical energy on calculus coating the teeth, generating cavitations in the fluid/liquid and causing the calculus to fracture, loosen and break away from the teeth enamel.

(23) In accordance with another example, there is also provided a dental treatment tray of a dental treatment applicator that can also include permanent magnets adhered to walls of the applicator by a suitable biocompatible adhesive or embedded inside the walls so that to create a magnetic field inside the well defined by the walls of the tray and thus improve the teeth whitening process carried out by ions of the active or whitening agent.

(24) In accordance with another example, there is also provided a method for teeth whitening treatment including applying to a dental arcade and active whitening agent, which could be in liquid or gel aggregate state, and activating the active agent by applying at least one form of energy selected from a group of forms of energy including light energy, electrical energy and acoustic-mechanical energy.

(25) In accordance with another example, there is also provided applying a permanent magnetic field to the dental arcade while concurrently applying to the active agent electrical energy.

(26) In accordance with another example the dental treatment applicators can also include a fluid conducting network configured for flushing and rinsing of the dental arcade being treated, removing fluid by suction or drying and soothing by blowing ambient air over the teeth and/or gums.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIGS. 1A and 1B are perspective-view and block view simplified illustrations of an apparatus according to an example;

(3) FIG. 1C is a cross-section view simplified illustration through a harness of FIGS. 1A and 1B;

(4) FIGS. 1D and 1E are perspective-view and exploded view simplified illustrations of another example;

(5) FIG. 2 is a perspective-view simplified illustration of still another example of a dental treatment dental treatment applicator;

(6) FIG. 3 is a perspective-view simplified illustration of another example of a dental treatment applicator;

(7) FIG. 4 is a perspective-view simplified illustration of yet another example of a dental treatment applicator;

(8) FIGS. 5A, 5B and 5C are perspective-view simplified illustrations of other examples of a dental treatment applicator;

- (9) FIG. 6 is a perspective-view simplified illustration of yet another example of pairing of dental treatment applicators of FIGS. 1A and 1B;
- (10) FIG. 7 is a perspective exploded-view simplified illustration of the example of FIG. 6;
- (11) FIGS. 8A and 8B are perspective-view simplified illustration of still other examples of a dental treatment applicator;
- (12) FIGS. 9A and 9B are perspective-view simplified illustrations of other examples of a dental treatment applicator; and
- (13) FIG. 10 is a perspective-view simplified illustration of still another example of a dental treatment applicator.

DETAILED DESCRIPTION

(14) The following description of the preferred embodiment(s) is merely exemplary in nature and is in no way intended to limit the invention, its application, or uses.

(15) The description of illustrative embodiments according to principles of the present invention is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. In the description of embodiments of the invention disclosed herein, any reference to direction or orientation is merely intended for convenience of description and is not intended in any way to limit the scope of the present invention. Relative terms such as “lower,” “upper,” “horizontal,” “vertical,” “above,” “below,” “up,” “down,” “top” and “bottom” as well as derivative thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing under discussion. These relative terms are for convenience of description only and do not require that the apparatus be constructed or operated in a particular orientation unless explicitly indicated as such. Terms such as “attached,” “affixed,” “connected,” “coupled,” “interconnected,” and similar refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. Moreover, the features and benefits of the invention are illustrated by reference to the exemplified embodiments. Accordingly, the invention expressly should not be limited to such exemplary embodiments illustrating some possible non-limiting combination of features that may exist alone or in other combinations of features; the scope of the invention being defined by the claims appended hereto.

(16) As used throughout, ranges are used as shorthand for describing each and every value that is within the range. Any value within the range can be selected as the terminus of the range. In addition, all references cited herein are hereby incorporated by reference in their entireties. In the event of a conflict in a definition in the present disclosure and that of a cited reference, the present disclosure controls.

(17) In the current disclosure all dental treatment applicators described and illustrated as single dental arcade applicators can be fitted on either one of the upper and lower human dental arcades and the dental treatment trays can be designed to be paired with a same or different dental treatment tray so that both dental treatment trays can be fitted at the same time (concurrently) onto their respective upper and lower dental arcades thereby treating both dental arcades concurrently. Other features, as will be explained in greater detail below, include accelerating teeth whitening treatment using, for example, electric current, Light Emitting Diodes (LED), piezoelectric elements and/or a magnetic field and calculus removal employing primarily piezoelectric elements.

(18) As shown in FIG. 1A dental treatment apparatus **100** could include a dental treatment applicator or tray **100-1** and a portable unit **118** that will be described in greater detail below. FIGS. 1A and 1B are perspective-view and block diagram simplified illustrations of an example, an upper dental arcade (FIG. 1A) and a lower dental arcade (FIG. 1B) dental treatment applicators or trays **100-1/100-2** can each be trough-shaped, curved in a form of a horseshoe so that to be configured to accommodate an upper or lower human dental arcade and an active agent. When paired, a pair of dental treatment applicators or trays **100-1/100-2** can be configured to accommodate both upper

and lower dental arcades concurrently.

(19) Dental treatment applicator or tray **100-1** (FIG. 1A) and **100-2** (FIG. 1B) could include a back curved wall **102** and a front curved wall **104** each integrally attached to opposite sides of floor **106** (FIG. 3) and together defining a well **112** configured to hold a substance such as, for example, an active or whitening agent (i.e., oxidizing chemical agent), which could be in liquid or gel aggregate state, with a suitable electric conductivity. Such as a suitable conductive active agent could be a teeth whitening liquid or gel with electric conductivity exceeding for example 200-400 microsiemens/cm. Use of such an active agent negates the need for an additional activating solution such as those used by the existing dental whitening devices. Walls **102/104** can also include energy-applying elements **138**. Energy-applying elements **138** can be configured to emit in course of dental whitening treatment energy such as, for example, light energy from light emitting diodes (LED) **802** (FIGS. 8A and 8B) or electric energy such as, for example, applied by electric current electrodes **140/140-1** (FIGS. 6 and 7) as will be explained in greater detail below. Energy-applying elements **138** could be configured and commutated or controlled to address treatment of groups of teeth or individual teeth. The existing teeth bleaching or whitening devices provide a uniform treatment throughout a dental arcade and are not designed to provide varying degrees of treatment to individual teeth within a human dental arcade.

(20) Since energy-applying elements **138** are employed to activate and/or accelerate the suitable conductive active or whitening agent, energy-applying elements **138** located on walls **102** and **104** do not have to come in contact with the teeth. In particular electric current flowing through the active agent from electrodes **140/140-1** could cause a redox reaction facilitating the teeth bleaching process.

(21) A rim **114** can be optionally thickened and optionally covered with a resilient material so that when dental treatment applicator **100/100-1** is placed over and accommodates a dental arcade, rim **114** forms a seal with the gums and prevents leakage of the active agent and in particular of the active agent with suitable electric conductivity out of well **112**. Optionally, rim **114** can be thickened and configured to extend beyond opening **140** up to a point of complete blockage, if desired, of opening **170**. Dental treatment applicator **100/100-1** can be made of any biocompatible material, for example, such as clear plastic or silicone so that to facilitate view of the level of an active agent inside well **112**.

(22) Additionally and optionally, dental treatment applicators or trays **100-1/100-2** can also include a fluid conducting network similar to network **520** (FIG. 5) for flushing and rinsing the dental arcade being treated and others.

(23) Dental treatment applicator or tray **100-1** could be connected via a harness **116** to a portable unit **118** that can include one or more of an alternating current (AC) or direct current (DC) power source **120**, a DC-current generator **124**. A computer or a processor **122**, a suction/vacuum pump **126**, a liquid reservoir **128**, a gel reservoir **130**, an ambient air fan **132** and a waste reservoir **134** could also be included in portable unit **118**. Power source **120** could be, for example, a standard electrical AC grid outlet or a battery configured to supply energy-applying elements **138** and DC-current generator **124**. DC-current generator **124** could be, for example, an alternating current (AC) rectifier or a battery. Some of the elements listed could be optional and not necessarily are included in each of the portable units **118**. Active or whitening agent in liquid or gel aggregate state could be respectively stored and dispensed from liquid reservoir **128** or gel reservoir **130**. Alternatively, the active agent could be manually filled in into well **112**.

(24) Portable unit **118** could also include one or more optional valves **160** configured to control flow of fluids such as the suitable active agents, water and ambient air through harness **116** conduits **150**. Valves **160** can be in communication with and controlled by computer or processor **122**.

(25) Energy-applying elements **138** can be configured to communicate with one or more of computer **122**, power source **120** and/or DC current generator **124**.

(26) As shown in FIG. 1C, which is a cross-section view simplified illustration through harness **116** of FIG. 1, harness **116** can also include one or more of electrical communication lines **120**, data communication lines **132**, conduits **150**

(27) Dental treatment applicator or tray **100-2** (FIG. 1B) could be paired (FIGS. 6 and 7) with dental treatment applicator **100-1** (FIG. 1A) and could be configured to connect directly to portable unit **118** via harness **116-1** or indirectly via dental treatment applicator or tray **100-1** as will be explained in greater detail below. Dental treatment applicator or tray **100-1** can be connected to harness **116** by one or more electrical and data communication connectors (not shown) such as, for example, a USB connector or any other suitable electrical and data communication connector known in the art and one or more fluid conduit connectors.

(28) As illustrated in FIGS. 1D and 1E, which are perspective-view and exploded view simplified illustrations of another example of apparatus **100**, where portable unit **118**, is attached to dental treatment applicator or tray **100-1**. In a basic implementation, portable unit **118** that could serve as a handle supporting a convenient handling of apparatus **100** and include one or more of alternating current (AC) or direct current (DC) power source **120**. In some examples, portable unit **118** can be configured to implement a specific dental treatment protocol and could be replaced with another portable unit **118** when a different dental treatment protocol is desired.

(29) Alternatively and optionally, portable unit **118** can include one or more disposable components such as liquid reservoir **128**, gel reservoir **130** configured to supply the liquid or gel into well **112**, waste reservoir **134** and power source **120** (FIG. 1A). As described above, portable unit **118** can serve as a handle and be attached to either one of dental treatment applicators or trays **100-1** and **100-2** or to both applicators **100-1** and **100-2**. Portable unit **118** can be designed so that the user can close his or her lips comfortably around portable unit **118** during the treatment session.

(30) In another example, illustrated in FIG. 2, which is a perspective-view simplified illustration of an example of a dental treatment applicator, dental treatment applicator or tray **200** can include piezoelectric elements **202** distributed along walls **102/104** at the level of the gums. Piezoelectric elements **202** can protrude from walls **102/104** so that to come into contact with the gums when dental treatment applicator **200** is applied to the dental arcade.

(31) When activated, piezoelectric elements **202** can vibrate at a frequency of between 20 kHz and 36 kHz and in some examples between 20 kHz and 100 kHz and a power range between 0.02 to 2.0 watt per element, operative to apply a massaging effect to the gums without the abrasive action commonly applied by apparatuses such as a tooth brush. The massaging effect can temporarily alleviate any discomfort or irritation brought about by active agents and/or oxidation process inside well **112**. Alternatively and optionally, piezoelectric elements **202** can be located along and protrude from walls **102/104** (e.g., at the level of the gums) so that not to come into contact with the gums when dental treatment applicator **200** is applied to the dental arcade (A layer of proper bio-compatible electrically insulating material could cover the piezo elements.). In this configuration, when activated, piezoelectric elements **202** can vibrate and assist in uniform distribution and mixing of the active agents throughout well **112** to accelerate their teeth whitening activity. The mixing activity of piezoelectric elements **202** can also replenish the active agents in areas where the active agent activity/concentration has been reduced by, for example, being diluted by saliva.

(32) Additionally and optionally, piezoelectric elements **202** can vibrate at a frequency of between 20 kHz and 36 kHz and in some examples between 20 kHz and 100 kHz and a power range between 0.02 to 2.0 watt per element that can bring about superficial debridement of one or more gum cell layers thus stimulating inflamed gum tissue re-growth and healing.

(33) FIG. 3, which is a perspective-view simplified illustration of yet another example of a dental treatment applicator. Applicator or tray **300** can be used as a dental scaler for calculus removal dental treatment. As illustrated in FIG. 3, piezoelectric elements **302** distributed along and protrude from walls **102/104** (e.g., at the level of the teeth) so that to come into contact with the teeth when

dental treatment applicator or tray **300** is applied to the dental arcade.

(34) When activated at a frequency of between 20 kHz and 36 kHz and in some examples between 20 kHz and 100 kHz and a power range between 0.02 to 2.0 watt per element, piezoelectric elements **302** can act as dental scalers by effecting acoustic-mechanical energy on calculus coating the teeth, generating cavitations in the surrounding fluid or liquid that in turn effect shock waves in the liquid causing the calculus to fracture, loosen and break away from the teeth enamel.

(35) Cavitations induced by piezoelectric elements **302** can also enhance the effect of chemical reactions mainly because of the high energy created by the temperatures and pressure emitted by the large number of individual cavitation bubble implosions in the calculus and liquid.

(36) In the presents of liquid, activation of piezoelectric elements **302** can generate vibrations inducing turbulent movement of the liquid that can help to remove the plaque from the tooth's surface and to flush out gum pockets. This activity can complement the cavitations-induced calculus removing activity. The turbulent movement of the liquid or gel could also be used to maintain a homogenous concentration of the active or whitening agent in well **112** and in some examples the piezoelectric elements **302** could be activated concurrently with the application of DC-current.

(37) Dental treatment applicator **300** can also include multi-purpose apertures **502** (FIG. 5) that can be employed to flush well **112** with a liquid such as water stored in optional liquid reservoir **128** or gel reservoir **130** (FIG. 1A) during or after the calculus removal treatment. Flushing could also be employed to cool the teeth and gums from heat generated by piezoelectric elements **302**. The flushing liquid can be removed by suction in the manner explained below.

(38) Reference is now made to FIG. 4, which is a perspective view simplified illustration of another example of a dental treatment applicator. As described in Water Electrolysis under a Magnetic Field by T. Iida et al. (Journal of the Electrochemical Society 2007, vol. 154, .sup.no8, [Note(s): E112-E115]), the energy efficiency of water electrolysis is considerably improved under a high magnetic field. Hence, application of electrolysis of the water based active or whitening agents (i.e., oxidizing chemical agents) under magnetic field conditions can intensify the process and speed it along, shortening the chair time of the subject being treated. As mentioned above, treating both dental arcades concurrently also contributes to shortening the chair time of the subject by at least halving the dental treatment time.

(39) Dental treatment applicator **400** can also include permanent magnets **402/404** adhered to walls **102/104** by a suitable biocompatible adhesive or embedded inside walls **102/104** so that to create a magnetic field inside well **112** and thus improve the teeth whitening process carried out by, for example, by application of DC-current through electrodes **140/140-1**.

(40) In some situations, employing permanent magnets in dental treatment applicator **400** can also simplify the treatment by effecting a magnetic field in well **112** directing the movement of free ions in liquid in well **112** thus negating the need for application of an electrical current to DC electrodes **140/140-1** (FIGS. 6, 7 and 8B).

(41) Reference is now made to FIGS. 5A, 5B and 5C, which are perspective-view simplified illustrations of still another example of a dental treatment applicator. As shown in FIG. 5A, dental treatment applicator or tray **500** can also include a fluid conducting network **520** configured, for example, for flushing and rinsing of the dental arcade being treated. The fluid conducting network **520** can include one or more multi-purpose apertures **502** in walls **102/104** that communicate with one or more common passages **504**, depicted in FIG. 5A by phantom lines, inside and along walls **102/104** and floor **106** and communicate in turn with one or more liquid reservoir **128**, gel reservoir **130** and waste reservoir **134** in portable unit **118** via dedicated conduits **150** (FIG. 1C) inside harness **116** or portable unit **118**.

(42) The fluid conducting network **520** and multi-purpose apertures **502** can act as nozzles to supply chemical oxidizing or active agents stored in liquid reservoir **128** or gel reservoir **130** (FIG. 1A), via one or more conduits **150** (FIG. 1C) in harness **116**, through common passages **504** and

into well **112**. The amount of chemical oxidizing or fluorinating agents and rate of supply into well **112** can be controlled via one or more valves **160** (FIG. **1C**) by computer or processor **122** (FIG. **1A**) in accordance with a predetermined protocol.

(43) Additionally or alternatively and optionally, the fluid conducting network **520** multi-purpose apertures **502** can act as intake ports, communicating with suction/vacuum pump **126** (FIG. **1A**) via one or more conduits **150** (FIG. **1C**) in harness **116** and common passages **504** to support rapid evacuation of well **112** from all liquid material into waste reservoir **134** (FIG. **1A**). To avoid formation of vacuum in well **112**, evacuation of liquid material could be immediately or concurrently followed by introduction of ambient air into well **112** by ambient air fan **132** (FIG. **1A**) via one or more conduits **150** (FIG. **1C**) in harness **116**, common passages **504** and through multi-purpose apertures **502**.

(44) Additionally or alternatively and optionally, the multi-purpose apertures **502** of the fluid conducting network **520** could be employed to flush well **112** with a liquid such as water stored in liquid reservoir **128** or gel reservoir **130** (FIG. **1A**), via one or more conduits **150** (FIG. **1C**) in harness **116**, through common passages **504** and into well **112**. Flushing could also be employed to cool the teeth and gums from heat generated by energy-applying elements **138** when activating the active agents (i.e., oxidizing chemical agents) during treatment. The flushing liquid can be removed by suction in the manner explained above.

(45) In another example depicted in FIG. **5B**, dental treatment applicator **550** can also include two or more resilient partitions **552**, positioned in pairs, distributed along walls **102/104** so that each partition on one of walls **102/104** is opposite its paired partition on the opposite wall **102/104** and normal to the plane of floor **106**. Partitions **552** apportion walls **102/104** into segment pairs, each segment on one of walls **102/104** opposite its paired segment on the opposite wall **102/104**. Each segment can include at least one DC current electrode and at least one multi-purpose aperture **502**. The frequency of location of DC-electrodes and multi-purpose apertures could be configured to support treatment of each individual tooth. The electrodes are positioned so that not to contact adjacent teeth when the applicator is applied to the dental arcade.

(46) In this example, computer or processor **122** (FIG. **1A**) could be configured to control location and rate of suitable conductive active or whitening agents (e.g., oxidizing chemical agents with an electric conductivity higher than 200-400 microsiemens/cm) introduction and activation during treatment in accordance with a predetermined protocol preset by a user. Partitions **552** can be located in between two adjacent teeth such that each segment accommodates at least a single tooth, thus each tooth can be treated individually. In the case of teeth whitening for example, individual tooth treatment can compensate for uneven distribution of staining or discoloration of various teeth. Computer or processor **122** can be configured to control and adjust treatment of each individual tooth based on signals received from sensors in dental treatment applicator such as, for example, from an optic sensor **580** as will be explained further below.

(47) The location of the fluid conducting network **520** multi-purpose apertures **502** along the height of walls **102/104** (i.e., the axis normal to plane of floor **106**) can be determined and set for a specific treatment. As shown in FIG. **5C**, multi-purpose apertures **502** can be distributed along the gum line only. This example can be employed, for example, for gum flushing and cooling purposes following a dental treatment session.

(48) Dental treatment applicator **570**, depicted in FIG. **5C** can also include at least one sensor selected from a group of sensors including a temperature sensor **572**, a liquid level gauge **576** such as, for example, a capacitance sensor, an oxygen sensor **578**, an optic sensor **580** and a pH meter **582**. The sensors can be configured to provide computer or processor **122** (FIG. **1A**) with feedback including real-time treatment parameters regarding ongoing dental treatment. Computer or processor **122** could be configured to adjust treatment process based on treatment parameters received from sensors **572/576/578/580** and **582** by varying location and rate of active agents (e.g., oxidizing chemical agents) introduction and activation, stopping treatment altogether, evacuating

well **112** from any material, flushing well **112** with flushing and/or cooling liquid or any other adjustment deemed necessary in accordance with the data received.

(49) For example, optic sensor **580** can be a color verification sensor that can monitor changes in the color of a single tooth or multiple teeth and signal computer **122** (FIG. **1**) when a desired level of whiteness has been achieved.

(50) Alternatively or additionally and optionally when employing, for example, a dental treatment applicator such as that depicted in FIG. **5B**, several optic sensors **580** can be configured to monitor color of individual teeth within a segment of a dental arcade or throughout a full dental arcade and provide computer **122** with information regarding the color of the individual teeth so that computer **122** can analyze color differences between various teeth based on signals received from optic sensor **580** and vary individual dental treatment accordingly.

(51) Real-time treatment parameters can include one or more parameters selected from a group of parameters including temperature inside well **112**, capacitance between the electrodes, impedance between the electrodes, level of activity (i.e., level of oxidation) of active agents, concentration of active agents inside well **112**, color of teeth, level of liquid inside well **112** and others.

(52) Reference is now made to FIGS. **6** and **7**, which are perspective-view simplified illustrations of examples of pairing of dental treatment applicators or trays **100-1** and **100-2**. As shown in FIG. **6**, dental treatment applicators **100-1/100-2** can be paired and configured so that to comfortably accommodate both upper dental arcade and lower dental arcade concurrently thus at least halving the required dental treatment time (i.e., shortening dental treatment time by at least 50 percent).

(53) Additionally and optionally, dental treatment applicators or trays **100-1/100-2** can each be operative to apply a different dental treatment to its corresponding dental arcade thus supporting application of different treatment protocols to different dental arcades concurrently negating the need for separate treatment times for each treatment protocol.

(54) FIG. **7**, which is an exploded perspective-view simplified illustration of paired dental treatment applicators or trays **100-1/100-2** of FIG. **2**, depicts one or more electrical and data communication connectors **702/702-1** comprising male **702** and female **702-1** components, each located, for example but not necessarily, on floor **106** of each of the paired dental treatment applicators or trays **100-1/100-2**. Electrical and data communication connector **702-1/702-2** facilitates electrical and data communication between dental treatment applicators **100-1/100-2** negating the need for dental treatment applicator **100-2** to connect directly to portable unit **118** and obviating harness **116-1** (FIG. **1**).

(55) Floors **106** of paired dental treatment applicators or trays **100-1/100-2** are configured to be removably attached to each other by attachments **704** facilitating easy attachment and detachment of dental treatment applicators **100-1/100-2** to and from one another. Attachments **704** can be, for example, strips of a mild adhesive allowing detachment of paired dental treatment applicators or trays **100-1/100-2** from one another by exertion of mild force or a fabric hook-and-loop fastener such as Velcro® (Velcro®, Amsterdam, The Netherlands). Alternatively and optionally, attachments **704** can be relatively weak magnets allowing detachment of dental treatment applicators **100-1/100-2** from each other by exertion of mild force. Attachments **704** are depicted on dental treatment tray **100-1** of FIG. **7** by phantom lines.

(56) As known in the art, both fluorination and teeth whitening procedures benefit from ionization by electrolysis of fluorinating and oxidizing agents by running an electrical current therethrough. Additionally, agents such as peroxides undergo oxidation in the presence of moisture, proteinaceous material (oral tissues), light and/or heat. Hence, heat and light application to the active agents, which could be in liquid or gel aggregate state (i.e., oxidizing chemical agents) can activate the process, intensify it and speed it along, shortening the chair time of the subject being treated. As mentioned above, treating both dental arcades concurrently also contributes to shortening the chair time of the subject by at least halving the dental treatment time.

(57) Reference is now made to FIG. **8A**, which is a perspective-view simplified illustration of an

example of a dental treatment applicator as depicted in FIG. 8A, dental treatment applicator or tray **800** energy-applying elements **138** of FIGS. 1A and 1B can be Light Emitting Diodes (LEDs) **802**. (58) LEDs **802** can emit an effective level of actinic light (e.g., UV light) and activate active agents such as whitening compositions or fluorinating compositions introduced into well **112** to effect rapid tooth whitening or fluorine uptake by the teeth. Additionally, LEDs **802** also generate heat further intensifying the process, speeding it along and shortening the chair time of the subject being treated.

(59) Referring back to FIGS. 4, 6, 7 and 8B, energy-applying elements **138** (FIGS. 1A and 1B) can be DC current electrodes **140/140-1** (FIGS. 4, 6, 7 and 8B) positioned on the inside surfaces of walls **102** and **104** and communicating with DC current generator **124** and computer **122** (FIG. 1A). The polarity of DC current electrodes **140** is opposite the polarity of electrodes **140-1**, i.e., if DC current electrodes **140** have a negative polarity (cathode), electrodes **140-1** have a positive polarity (anode) and vice versa, maintaining a voltage therebetween and generating a current when an active agent with suitable electrical conductivity, for example, a teeth whitening gel or liquid, is placed in well **112**, in contact with both electrodes **140** and **140-1**.

(60) When applying DC current energy to the dental arcade via DC current electrodes **140/140-1**, in the presence of a suitable conductive whitening agent, fluid or gel, the current generated between the electrodes activates the oxidizing agent thus enhancing the agent's activity, accelerating the treatment. Additionally, DC current energy applied by DC current electrodes **140/140-1** generates heat within the active agent which could be a fluid or gel, further accelerating the treatment and shortening chair time of the subject being treated. Electrodes **140** and **140-1** are placed on walls **102** and **104** so that when treatment trays **100-1/100-2** are placed onto the respective dental arcades electrodes **140/140-1** contact only the fluid or gel with suitable electric conductivity and do not contact the teeth themselves. In such a configuration when using a suitable conductive gel that can be a gel having, for example, conductivity exceeding 200-400 microsiemens/cm, no additional activating agent or solution is required.

(61) Alternatively and optionally and as depicted in FIG. 8B, dental treatment applicator **800** could include both DC current applying electrodes **140/140-1** and Light Emitting Diodes (LEDs) **802**. LED's **802** can be activated to generate photoactinic light and generate heat to sensitize and begin activation of the whitening composition or fluorinating composition followed by electrodes **140/140-1** activation so that to bring about full activation of the active or whitening agent composition without further elevation of temperature of the active agent compositions and/or dental arcade (teeth and/or gums) and prevent discomfort to the subject being treated. In this configuration, LED's **802** and electrodes **140/140-1**, controlled by computer or processor **122** that can address individually each tooth or segment of the dental arcade.

(62) Alternatively and optionally, and as depicted in FIG. 9A, which is a perspective-view simplified illustration of another example of a dental treatment applicator, dental treatment applicator **800** one or more electrodes **902** can be in a form of an electrically conductive strip and placed along the curved surface of walls **102** and/or **104**. In another configuration, shown in FIG. 9B, continuous electrodes in the form of electrically conductive strips **902/904** are placed on walls **104** and **102** respectively and in contact with only the fluid or gel with suitable electric conductivity and do not contact the teeth themselves. During treatment, electrodes **902/904** can be electrically connected to opposite electrical poles so to cause a redox reaction facilitating the teeth bleaching process.

(63) As shown in FIG. 10, which is a perspective-view simplified illustration of still another example of a dental treatment applicator energy-applying elements **138** of FIGS. 1A and 1B can be a plurality of DC current-applying elements **1002/1002-1**.

(64) The polarity of DC current-applying elements **1002** is opposite the polarity of DC current-applying elements **1002-1**, i.e., if DC current-applying elements **1002** have a negative polarity (cathode), DC current-applying elements **1002-1** have a positive polarity (anode) and vice versa,

maintaining a voltage therebetween and generating a current when an electrolytic chemical agent is placed in well **112**, in contact with both DC current-applying elements **1002** and **1002-1**.

(65) Additionally and optionally, DC current-applying elements **1002-2/1002-1** can be individually supplied by DC current energy generator **124** and individually controlled by computer or processor **122** so that the DC current between pairs of DC current-applying elements **1002-2/1002-1** can vary at various locations along the dental arcade being treated by the activated whitening gel with suitable electric conductivity and apply treatment to one or more segments of the dental arcade or to individual teeth without being in contact with the teeth. For example, dental treatment can vary from treatment at discrete locations through regional activation (e.g., provide a gradient of level of activity along the dental arcade) to full dental arcade treatment of fluorination or whitening treatment to the teeth.

(66) It will be appreciated by persons skilled in the art that the present method and apparatus are not limited to what has been particularly shown and described hereinabove. Rather, the scope of the method and apparatus includes both combinations and sub-combinations of various features described hereinabove as well as modifications and variations thereof which would occur to a person skilled in the art upon reading the foregoing description and which are not in the prior art.

Claims

1. A dental treatment apparatus comprising: a first dental treatment device comprising a first wall and a floor extending from the first wall, the first wall and the floor configured to accommodate an upper or lower dental arcade, the first wall having an inner surface; a plurality of first electrodes located on the inner surface of the first wall; a plurality of first light sources located on the inner surface of the first wall; and an optical sensor configured to monitor a color of one or more teeth; wherein the plurality of first electrodes and the plurality of first light sources are positioned in an alternating manner along the first wall.
2. The dental treatment apparatus according to claim 1 further comprising a processor operably coupled to the plurality of first electrodes, the plurality of light sources, and the optical sensor, the processor configured to control the plurality of first electrodes and the plurality of first light sources based on a signal received from the optical sensor.
3. The dental treatment apparatus according to claim 1 wherein the first wall extends from a first end to a second end, and wherein the plurality of first electrodes and the plurality of first light sources are arranged in an alternating manner along the first wall as the first wall extends from the first end to the second end.
4. The dental treatment apparatus according to claim 1 wherein the plurality of first electrodes and the plurality of first light sources are located on the first wall at a first height above the floor.
5. The dental treatment apparatus according to claim 1 wherein each of the first electrodes is spaced apart from the first electrodes that are adjacent thereto by a gap.
6. The dental treatment apparatus according to claim 5 wherein the first wall terminates in a distal edge, and further comprising a plurality of piezoelectric elements located on the inner surface of the first wall within the gaps between the first electrodes.
7. The dental treatment apparatus according to claim 6 wherein at least a portion of each of the piezoelectric elements is located on the inner surface of the first wall at an elevation above the floor that is between the distal edge of the first wall and an uppermost portion the first electrodes.
8. The dental treatment apparatus according to claim 7 wherein each of the piezoelectric elements protrudes from the inner surface of the first wall.
9. The dental treatment apparatus according to claim 1, wherein the plurality of first electrodes are at least partially embedded within the first wall so that an outer surface of each of the plurality of first electrodes.
10. The dental treatment apparatus according to claim 1, further comprising a portable unit

detachably coupled to the first dental treatment device, the portable unit comprising a housing that contains a power source and a processor that are operably coupled together and to the plurality of first electrodes.

11. The dental treatment apparatus according to claim 1 further comprising a second dental treatment device detachably coupled to the first dental treatment device, the second dental treatment device comprising: a first wall and a floor extending from the first wall, the first wall of the second dental treatment device configured to accommodate an upper or lower dental arcade; and a plurality of second electrodes located on the first wall of the second dental treatment device.

12. The dental treatment apparatus according to claim 11 wherein the first and second dental treatment device comprise bottom surfaces that face one another when the first and second dental treatment device are coupled together, a first electrical and data communication connector located on the bottom surface of the first dental treatment device and a second electrical and data communication connector located on the bottom surface of the second dental treatment device, and wherein the first and second electrical and data communication connectors mate with one another to electrically couple the first and second dental treatment devices together.

13. The dental treatment apparatus according to claim 12 further comprising a portable unit detachably coupled to the first dental treatment device, the portable unit comprising a housing that contains a power source and a processor that are operably coupled together, and wherein the power source and the processor of the portable unit power the plurality of first electrodes of the first dental treatment device and the plurality of second electrodes of the second dental treatment device.

14. A dental treatment apparatus comprising: a first dental treatment tray comprising a first wall, a second wall that is spaced apart from the first wall, and a floor extending between the first and second walls, the first and second walls and the floor collectively defining a well configured to accommodate an upper or lower dental arcade, each of the first and second walls having an inner surface that faces the well; and a first electrode strip located on the inner surface of the first wall and a second electrode strip located on the inner surface of the second wall, the first and second electrode strips configured to apply electric current to an active agent disposed within the well.

15. The dental treatment apparatus according to claim 14 wherein the inner surface of the first wall is concave and the inner surface of the second wall is convex, the first electrode strip having a concave exposed surface and the second electrode strip having a convex exposed surface, the first and second electrode strips being aligned with one another, wherein the first electrode strip is at least partially embedded within the first wall so that an outer surface of the first electrode strip that faces the well is exposed, and wherein the second electrode strip is at least partially embedded within the second wall so that an outer surface of the second electrode strip that faces the well is exposed.

16. A method of treating an oral cavity comprising: providing a dental treatment tray comprising: a first wall, a second wall that is spaced apart from the first wall, and a floor extending between the first and second walls, the first and second walls and the floor collectively defining a well, each of the first and second walls having an inner surface that faces the well; at least one electrode located on the inner surface of each of the first and second walls; and at least one light source located on the inner surface of at least one of the first and second walls; placing an amount of an active agent into the well so that an exposed surface of the electrodes on the first and second walls and an exposed surface of the light source are in contact with the active agent; positioning the dental treatment tray into an oral cavity so that an upper or lower dental arcade is positioned within the well; activating the at least one light source to generate light and heat that is applied to the active agent in the well; and after the at least one light source has been activated for a predetermined period of time, activating the electrodes to generate an electric current that is applied to the active agent in the well.
